

Observing Students in Classroom Settings: A Review of Seven Coding Schemes

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Abstract. A variety of coding schemes are available for direct observational assessment of student classroom behavior. These instruments have been used for a number of assessment tasks including screening children in need of further evaluation for emotional and behavior problems, diagnostic assessment of emotional and behavior problems, assessment of classroom ecology in the formulation of academic interventions, and monitoring the progress of medical, psychosocial, and academic interventions. Although this method of behavioral assessment has a high degree of face validity, it is essential to consider the psychometric properties of available coding schemes to select the appropriate instrument for a given assessment. This article reviews the structure, content, training requirements, and available psychometric properties of seven available direct observation codes. Recommendations for the use of each code and future directions for research in observational assessment are provided.

"You can observe a lot just by watchin'."

Yogi Berra

Systematic observation of student behavior is among the most common assessment methodologies utilized by school psychologists (Shapiro & Heick, 2004; Wilson & Reschly, 1996) and is viewed as one of the most objective and direct measurement tools available for the assessment of child behavior. This method traditionally has been used for a number of

assessment tasks including: (a) screening children in need of further evaluation for emotional and behavior problems, (b) diagnostic assessment of emotional and behavior problems, (c) assessing classroom ecology in the formulation of academic interventions, and (d) monitoring the progress of medical, psychosocial, and academic interventions.

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Information gathered via direct observation has a high degree of face validity; however, several factors may have a negative effect on the quality of these data. Merrell (1999) lists the following threats to the validity of behavioral observation: (a) poorly defined behavior categories, (b) low interobserver reliability, (c) observee reactivity, (d) situational specificity of target behaviors, (e) inappropriate code selection, and (f) observer bias. These threats may be minimized through the selection and use of well-validated instruments and adequate training in their use. Hintze (2005) provides specific guidelines for the selection of appropriate coding schemes. Because no observation code is appropriate for all situations, the selection of an appropriate observation coding scheme is an essential step in maximizing the validity of an observation-based assessment.

To select the observation code that is most appropriate for a given purpose, users should be aware of the available codes, their composition, their psychometric properties (see Hintze, 2005), and the amount of time required to learn each code. Although several resources exist that describe observation coding schemes (e.g., Hintze, Volpe, & Shapiro, 2002; Winsor, 2003), they include only a few measures and provide limited information regarding the psychometric properties of each measure. The purpose of this article is to provide a comprehensive review of observation coding schemes available to assess the academic behaviors of elementary school children. Although coding schemes exist to measure a wide variety of student behaviors in classrooms (e.g., academic behaviors, social behaviors, student-teacher interactions), this review focuses on codes measuring academic engagement given the central role this observable classroom behavior plays in the learning process (e.g., Greenwood, 1996). The ultimate goal is to assist school psychologists' with the selection of appropriate observation codes for use in their professional practice.

Selection of Observation Codes for Review

In keeping with our aforementioned purpose, we limited our review to coding schemes

easily obtained by school practitioners that include a measure of academic engagement and can be used without the need of a computer. Although excellent computer-based coding systems exist for assessing child academic behaviors such as the Eco-Behavioral Assessment Software System (Greenwood, Carta, Kamps, & Delquadri, 1993), the need of a portable computer and extensive training limits their widespread use by school practitioners. To be included in this review, a coding system must: (a) have been designed for use in elementary classrooms; (b) have a manual available from the author, or otherwise published, in a form to facilitate standardized use of the code; (c) assess academically related behaviors (e.g., academic engagement); and (d) allow for recording via a paper-and-pencil format (some available codes can be used only with portable technology, such as a laptop computer or PDA). A search of electronic databases (e.g., ERIC Assessment Clearinghouse, PsychInfo), Buros *Mental Measurements Yearbooks*, recent test catalogs, and attempts to contact developers yielded the following seven codes meeting our inclusion criteria: Academic Engaged Time Code of the Systematic Screening for Behavior Disorders (AET-SSBD; Walker & Severson, 1990), ADHD School Observation Code (ADHD-SOC; Gadow, Sprafkin, & Nolan, 1996), Behavioral Observation of Students in Schools (BOSS; Shapiro, 2004), Classroom Observation Code (COC; Abikoff & Gittelman, 1985), Direct Observation Form (DOF; Achenbach, 1986), State-Event Classroom Observation System (SECOS; Saudargas, 1997), and Student Observation System of the Behavioral Assessment System for Children-2 (SOS; Reynolds & Kamphaus, 2004).

For each systematic observation code included in this review, we report the following information: (a) the general purpose of the code, (b) the length of time required for training (as reported in research studies, these are likely overestimates for experienced school psychologists), (c) a list of behavior categories (with examples of target behaviors), and (d) a summary of published psychometric data. In addition, we address the strengths and limi-

tations of each code and offer recommendations for its appropriate use. Table 1 presents characteristics of each code, including availability, availability on hand-held computers, recording method (e.g., partial interval, momentary time sample), behavior categories, training requirements, and the typical length of single observations. Table 2 summarizes the psychometric properties of each code. Table 3 summarizes the strengths and limitations of each code and offers recommendations for use based on available data. In the following sections we review each of the seven systematic direct observation codes.

Academic Engaged Time Code (AET-SSBD)

The AET-SSBD is a component of Systematic Screening of Behavior Disorders (SSBD; Walker & Severson, 1990), a screening procedure for emotional and behavior disorders in elementary school children (Grades 1–6). The full SSBD system involves three gates: (a) teacher rank ordering of students in terms of externalizing and internalizing problems, (b) teacher ratings of adaptive behavior and critical events, and (c) systematic direct observations in multiple school settings (the AET-SSBD in classroom settings, and the Peer Social Behavior Code or PSB-SSBD in free-play settings). Leff and Lakin (2005) provide a review of the PSB in this issue. The AET-SSBD and PSB-SSBD are used in the third gate to verify teacher ratings. The AET-SSBD measures the amount of time a student spends engaged in academic material during independent seatwork (e.g., listening to the teacher, writing in a workbook). The total amount of time a student exhibits behavior consistent with the definition of academic engagement is recorded with a stopwatch. This value is divided by the overall time of observation (usually 15 minutes) and multiplied by 100 to compute academic engaged time. Typically data from two observations are averaged to obtain a stable academic engaged time score. As opposed to comparing this score to one or more randomly selected peers, in the AET-SSBD raw scores are converted to *T*-scores using normative data tables. Normative data for the SSBD-AET con-

sist of a sample of 1,300 first through sixth grade children from 16 school districts across six states. Normative tables in the SSBD manual are arranged according to gender and whether or not the child of interest has met the criteria for either internalizing or externalizing problems in the second stage of the SSBD. Training requirements have been reported to consist of as little as 4 to 6 hours for both the AET-SSBD and PSB-SSBD codes (Walker et al., 1990).

Psychometric properties. For the AET-SSBD, interobserver agreement is calculated by dividing the smaller score of two observers by the larger score. Mean interobserver agreement coefficients across five published studies have ranged from .95 (Walker et al., 1994) to .98 (Quinn, Mathur, & Rutherford, 1995). Coefficients for individual cases have typically ranged from the .80s to 1.00. Scores on the AET-SSBD have been shown to correlate significantly ($r = -.42$) with teacher ratings of externalizing behavior problems (Walker et al., 1988). Several studies have found scores on the AET-SSBD to accurately discriminate control children from those nominated as at-risk by teachers for externalizing problems (e.g., Quinn et al., 1995; Walker et al., 1990). Less consistent has been the ability of scores from the AET-SSBD to discriminate children nominated as at-risk for internalizing problems from controls (Walker et al., 1988; Walker et al., 1990).

Attention-Deficit Hyperactivity Disorder School Observation Code (SOC)

The ADHD-SOC (Gadow et al., 1996) was developed as both a screening measure and as a tool for evaluating the effects of interventions for children with attention-deficit/hyperactivity disorder (ADHD) and related disorders across school settings (classroom, lunchroom, playground). According to its developers, training for the ADHD-SOC should require approximately 20 to 25 hours. Categories are coded in 15-second intervals for 15 minutes. The following behaviors are coded using the partial interval method: (a) interference (e.g.,

Table 1
Characteristics of Reviewed Systematic Observation Codes

Code/Availability	Computerized?	Recording Method(s)	Behavior Categories	Training Requirements	Typical Length of Observation
Academic Engaged Time Code of the SSBD (AET-SSBD; Walker & Severson, 1990) www.sopriswest.com	No	Duration	1. Academic engaged time	Low	15 minutes
ADHD School Observation Code (ADHD-SOC; Gadow, Sprafkin, & Nolan, 1996) Available from: www.checkmateplus.com	No	Partial interval 15-second intervals	1. Interference 2. Motor movement 3. Noncompliance 4. Nonphysical aggression 5. Off-task	Moderate	15 minutes
Behavioral Observation of Students in Schools (BOSS; Shapiro, 2004) Computer version available from: www.sopriswest.com	Palm	Momentary/Time Sample Partial/Interval 15-second intervals	1. Active engaged time 2. Passive engaged time 3. Off-task motor 4. Off-task verbal 5. Off-task passive 6. Teacher directed instruction	Moderate	15 minutes
Classroom Observation Code (COC; Abikoff & Gittelman, 1985) This reference provides a detailed description of code and the observation protocol.	No	Partial interval Whole interval	1. Interference 2. Minor motor movement 3. Gross motor standing 4. Gross motor vigorous 5. Physical aggression 6. Verbal aggression 7. Solicitation of teacher 8. Off-task 9. Noncompliance 10. Out of chair behavior 11. Absence of behavior	High	32 minutes

(Table 1 continues)

(Table 1 continued)

Code/Availability	Computerized?	Recording Method(s)	Behavior Categories	Training Requirements	Typical Length of Observation
Direct Observation Form (DOF; Achenbach, 1986) www.aseba.org	No	Predominant activity sampling 4-point Likert scale for 97 problem items	1. On-task 2. Withdrawn-Inattentive 3. Nervous-Obsessive 4. Depressed 5. Hyperactive 6. Attention Demanding 7. Aggressive	Low	10 minutes
State-Event Classroom Observation System (SECOS; Saudargas, 1997) rsaudarg@utk.edu	No	Momentary/time sample Frequency recording 15-second intervals	States 1. School work 2. Looking around 3. Other activity 4. Social interaction with child* 5. Social interaction with teacher 6. Out of seat Events 1. Out of seat 2. Approach child 3. Other child approach 4. Raise hand 5. Calling out to teacher 6. Teacher approach	Moderate	20 minutes
Student Observation System (SOS; Reynolds & Kamphaus, 2004) www.agsnet.com	Windows Palm	Momentary/time sample 30-second intervals	Adaptive Behaviors 1. Response to teacher/lesson 2. Peer interaction 3. Works on school subjects 4. Transition movement	Unclear	15 minutes

(Table 1 continues)

(Table 1 continued)

Code/Availability	Computerized?	Recording method(s)	Behavior Categories	Training Requirements	Typical Length of Observation
		3-point Likert scale for 65 behavior items—some items permit scoring of whether the behavior was disruptive to the class	Problem Behaviors 1. Inappropriate movement 2. Inattention 3. Inappropriate vocalization 4. Somatization 5. Repetitive motor movements 6. Aggression 7. Self injurious behavior 8. Inappropriate sexual behavior 9. Bowel/bladder problems		

Note. Low (up to 10 hours), Moderate (between 11-25), (High > 25).

Table 2
Psychometric Properties of Reviewed Systematic Observation Codes

Code	Interobserver Agreement	Validity		Treatment Sensitivity	Normative Data
		Convergent	Discriminant		
AET-SSBD	Percent agreement averaged across 5 studies was .96	TRF Externalizing ($r = -.42$)	Discriminated children at-risk for externalizing problems from controls	No data available	1,300 children from 16 school districts in 6 states
			Discriminated children at-risk for internalizing problems from controls		Data available by gender, and by grouping of the SSBD (externalizing, internalizing, nonranked)
ADHD-SOC	Kappas = .77–.86 Kappas = .67–.80	IOWA ($rs = .06-.71$) ATRS ($rs = .11-.62$) Peer Conflict Scale ($rs = .38-.74$)	Discriminated children with emotional and behavioral disorders from children with learning disabilities	Sensitive to stimulant drug effects	Authors recommend use of classroom comparison children (observed in alternating 1-minute segments)
			Discriminated children with ADD from their nondisabled peers		
			Discriminated children with ADHD from their nondisabled peers		

(Table 2 continues)

(Table 2 continued)

Code	Interobserver Agreement	Validity			
		Convergent	Discriminant	Treatment Sensitivity	Normative Data
BOSS	Kappas = .93–.98	No published data available	Discriminated children with ADHD from their nondisabled peers	Sensitive to instructional manipulations	Authors recommend use of classroom comparison children (observed every fifth interval)
COC	Phi coefficients = .80–1.0 Phi coefficients = .55–.95 Phi coefficients = .40–.97 *Poor reliability for verbal aggression in some studies	No published data available	Discriminated children with ADHD from their nondisabled peers		
			Using multiple categories 80% of ADHD and typically developing children were correctly classified	Sensitive to psychosocial interventions and stimulant drug effects	Authors recommend use of classroom comparison children (observed in alternating 4-minute segments)
DOF	Average Pearson correlations across 4 studies was .90 for total behavior problems and .84 for on-task	TRF total behavior problems score ($r_s = -.26$ – $-.53$)	Discriminated boys referred for problem behaviors from typically developing boys matched for age, grade, and race	On-task, and the Nervous/Obsessive, and Depression scales have demonstrated sensitivity to a targeted prevention program	Authors recommend use of classroom comparisons observed in 10-minute blocks before and after 10-minute observation of target child.
	One way ICCs for 60 minutes of observation was .86 for total behavior problems and .71 for on-task	TRF school performance ($r_s = -.14$ – $.66$)			
	One way ICC for 10 minutes of observation was .85 for total behavior problems and .58 for on-task	TRF Adaptive functioning composite ($r_s = .48$ – $.72$)			

(Table 2 continues)

(Table 2 continued)

Code	Interobserver Agreement	Validity		Treatment Sensitivity	Normative Data
		Convergent	Discriminant		
SECOS	Average total agreement = .81	No published data available	Discriminated between children with behavior disorders and their nondisabled peers	No published data available	500 children in Grades 1 through 5
	Total agreement = .75–1.0		Discriminated between children with learning disabilities from their nondisabled peers		Children attended 1 of 10 elementary schools in East Tennessee
SOS	Unclear from available data	No published data available	Discriminated between children with ADHD and nondisabled children	No published data available	Authors recommend use of two or three randomly selected peer comparisons

IOWA = IOWA Conners Teacher's Rating Scale (Loney & Milich, 1982); ATRS = Abbreviated Teachers Rating Scale (Conners, 1973); TRF = Teacher Report Form (cf., Achenbach & Rescorla, 2001).

Table 3
Strengths, Limitations, and Recommendations for Reviewed Systematic Observation Codes

Code	Strengths	Limitations	Recommended Use
Academic Engaged Time-SSBD	Simple code to learn and use Low training requirements Strong reliability Strong support for discriminant validity	Narrow assessment of student behavior Need for updated normative data	Use as part of the SSBD for screening purposes Useful for measuring student engagement as part of a diagnostic assessment
ADHD-School Observation Code	Broad measurement of externalizing behaviors Known psychometric properties Sensitive to effects of treatment	Variable interobserver agreement Relatively low association with teacher ratings of hyperactivity	Assessment of externalizing problems Monitoring effects of interventions
BOSS	Specific assessment of active student engagement Some evidence for treatment sensitivity	More information needed regarding treatment sensitivity In general, limited information concerning psychometric properties	Describing the classroom behavior of children May be useful in assessment of externalizing behavior
COC	Strong support for discriminant validity Strong support for treatment sensitivity	Complex code that is a challenge to learn Data needed examining discrimination of children with ADHD from other affected populations	Screening and diagnosis of ADHD Monitoring the effects of interventions for ADHD
DOF	Easy to learn and use Broad assessment of externalizing and internalizing behaviors Integrated into a broad assessment system (ASEBA) Some evidence of treatment sensitivity Some evidence for utility in diagnostic assessments of behavior problems	More information needed with regard to the psychometric properties of scales other than total behavior problems More information needed regarding treatment sensitivity and discriminant validity Small normative sample	As part of the ASEBA for assessment of emotional and behavior problems

(Table 3 continues)

(Table 3 continued)

Code	Strengths	Limitations	Recommended Use
SECOS	Known accuracy Evidence for discriminant validity Low inference/descriptive categories Has been used on a wide age range (first grade–high school)	Normative group consists only of 500 students from East Tennessee No data available regarding treatment sensitivity	Assessment of externalizing problems if peer comparisons are utilized
SOS	Relatively broad assessment of both positive and negative student behaviors	Limited evidence of psychometric properties	None at this time

calling out when it is not appropriate to do so), (b) motor movement (e.g., getting out of seat without permission), (c) verbal aggression (e.g., cursing at another student), (d) symbolic aggression (e.g., taking another student's pencil), (e) object aggression (e.g., kicking a chair), and (f) off-task (e.g., looking out a window instead of completing an assignment). Non-compliance (e.g., ignoring verbal direction from teacher) is scored using the whole interval method. Other categories are coded in lunchroom and playground settings to assess appropriate and inappropriate social behaviors (see Leff & Lakin, 2005, for use of the ADHD-SOC in playground settings). When used as part of a comprehensive diagnostic assessment, the authors of the ADHD-SOC recommend selecting three or four peers to observe for comparison. Selected peers are observed with the target student in alternating 1-minute segments.

Psychometric properties. Interobserver agreement using the ADHD-SOC has been somewhat variable. For example, Nolan and Gadow (1994) reported kappa coefficients between .77 and .86 for the five classroom categories, with only the category of nonphysical aggression falling below .80. However, Gadow, Nolan, Sprafkin, and Sverd (1995) reported kappas at or below .80 for all five categories ($k = .67-.80$). Test-retest coefficients based on observations within a 2-week period were low to moderate (range = .27-.72) (Gadow et al., 1996). The association between teacher ratings of hyperactivity and relevant ADHD-SOC categories (motor movement and off-task) has been low and not statistically significant. However, significant associations between teacher ratings of hyperactivity and observed off-task behavior (though not motor movement) emerge when teacher ratings of negative behavior were controlled for statistically (r s between .46 and .48) (Gadow et al., 1996). Evidence for the convergent validity of the remaining categories of the ADHD-SOC (interference, noncompliance, nonphysical aggression) is more robust. Nolan and Gadow (1994) found moderate correlations between these categories and teacher ratings of aggression and emotional lability (range = .38-.74). In addition, the ADHD-SOC has been found to discriminate between children iden-

tified as having ADHD and their nonlabeled peers (Gadow et al., 1992). Finally, the treatment sensitivity of all but one ADHD-SOC category (nonphysical aggression) has been demonstrated in school-based studies of stimulant drug effects (Gadow, Nolan, & Sverd, 1992; Gadow, Nolan, Sverd, Sprafkin, & Paolicelli, 1990).

Behavioral Observation of Students in Schools (BOSS; Shapiro, 2004)

The BOSS was designed to assess student academic behavior in the classroom environment. According to its developer (the fourth author of this article), it should take between 10 and 15 hours of training to become proficient using the BOSS. The BOSS essentially measures levels of on- and off-task behavior. However, the BOSS divides on-task behavior into active engaged time (AET; coded when a student is actively engaged in academic responding; e.g., reading aloud, writing in a journal), and passive engaged time (PET; coded when a student is passively attending; e.g., listening to a teacher, looking at the blackboard while a teacher writes). Both AET and PET are scored using momentary time sampling at the beginning of each 15-second interval. During the remainder of each interval, the partial interval method is used to record the following off-task behavior categories: (a) off-task motor (motor activity not associated with the assigned academic task; e.g., leaving seat to throw a piece of paper in the trash can), (b) off-task verbal (utterances not associated with the academic task; e.g., talking to a peer about something other than the current assignment, humming), and (c) off-task passive (passive nonengagement; e.g., looking out the window). Using the BOSS, the observer codes the behavior of the target child for four out of every five intervals. On every fifth interval, the behavior of one of several preselected peers is coded on the same behaviors as the target child for comparison purposes. Finally, teacher-directed instruction (TDI) is coded using the partial-interval method. Scores on TDI estimate the amount of time a teacher is engaged in instruction. For example, TDI would be coded if the teacher

was lecturing to the class, but TDI would not be coded if he was grading papers at his desk. TDI, like peer comparison data, is scored on every fifth interval.

Psychometric properties. Reports of interobserver agreement for the BOSS have been consistently high. For example, in a study involving repeated measurement of three participants, Ota and DuPaul (2002) reported total agreement ranging between 90 and 100%. More recently, DuPaul et al. (2004) reported kappas ranging from .93 to .98 for observations in a large sample of children with ADHD and normal comparison children ($N = 136$). Although there are no data available supporting the convergent validity of the BOSS, there are some data supporting the ability of the BOSS to discriminate between children with ADHD and typically developing children. Specifically, DuPaul et al. (2004) found that PET and a composite of the three off-task scores of the BOSS significantly discriminated between children with ADHD who had academic problems and typically developing peers, whether the observations were conducted during instruction in mathematics or reading. Effect sizes for these variables ranged between $-.53$ and 1.25 .

Treatment sensitivity of the BOSS has been documented in a study investigating the efficacy of computer-aided instruction for three children with ADHD (e.g., Ota & DuPaul, 2002). In a multiple-baseline design across three participants, the BOSS categories of AET (ES between -2.91 and -13.01) and a composite of the three off-task scores (ES between 1.8 and 3.06) were found to be sensitive to manipulations in instructional modality (regular math instruction vs. working on a computer).

Classroom Observation Code (COC)

The COC (Abikoff & Gittelman, 1985) was designed to quantify the classroom behavior of children for diagnostic assessment for ADHD and for monitoring the effects of interventions designed to ameliorate the symptoms of ADHD. The COC is a relatively complex code consisting of 12 behavior categories. Abikoff, Gittelman-Klein, and Klein (1977) re-

ported that training for the code averaged 50 hours, and that only 5 of 8 advanced undergraduate and graduate student research assistants met the training criteria of 70% agreement at the end of training. Like the ADHD-SOC, the COC focuses exclusively on child behaviors. Categories of the COC are recorded in 15-second intervals using one of two sampling methods. The following discrete behaviors are scored using the partial interval method: (a) interference (e.g., calling out during a teacher lecture), (b) minor motor movement (e.g., twisting and turning while seated), (c) gross motor standing (e.g., out of seat and standing), (d) gross motor-vigorous (e.g., running or crawling across the classroom), (e) physical aggression (e.g., kicks or hits another child), (f) threat or verbal aggression-to children or -to teacher (e.g., curses at another child or teacher), and (g) solicitation of teacher (e.g., raises hand). The following behaviors are coded using the whole interval method: (a) off-task (e.g., plays with toy while the teacher is talking), (b) noncompliance (e.g., ignores verbal direction from teacher), and (c) out of chair behavior (e.g., out of seat when not appropriate to do so). Finally, if none of the aforementioned behaviors are noted in an interval, "absence of behavior" is coded. Observation sessions using the COC typically are 32 minutes in duration. A target child and a same gender teacher-nominated "normal" peer are observed for 16 minutes each, in alternating 4-minute blocks.

Psychometric properties. Reported interobserver agreement for the COC has been high. For example, Abikoff et al. (2002) collected interobserver agreement data for 10% of 1,893 observations, which yielded mean phi coefficients ranging from $.80$ to 1.00 . The discriminant validity of the COC is well documented (Abikoff et al., 1977; Abikoff, Gittelman, & Klein, 1980; Abikoff et al., 2002). For example, in a study of 502 pairs of children with ADHD and their classmates, Abikoff et al. (2002) found that all of the COC categories significantly discriminated between children with ADHD and their typically developing peers. The categories of off-task and interference have been found to be the most discriminating, correctly classifying 77% and

76.2% of cases, respectively (e.g., ADHD vs. “normal”). However, by combining the categories of interference, off-task, minor motor movement, gross motor movement, and solicitation, almost 80% of cases were correctly classified (Abikoff et al., 1980). The treatment sensitivity of the COC is well documented and has been used as a dependent measure in numerous studies of medical and psychosocial interventions for children with ADHD (e.g., Abikoff et al., 2004; Klein & Abikoff, 1997). Based on our review of the extant literature, no studies have investigated the convergent validity of the COC.

Direct Observation Form (DOF)

The DOF (Achenbach, 1986) was designed to obtain ratings of problem behaviors and on-task behavior directly observed in group settings, and is part of the Achenbach System of Empirically-Based Assessment (ASEBA; Achenbach & Rescorla, 2001). The DOF has been used in research studies across a number of school settings, including the classroom, lunchroom, and playground. Training for the DOF should take about 10 hours. Although each observation period is relatively brief (10 minutes), the developers of the DOF recommend that three to six observations be performed to gain a stable estimate of child behavior. During each observation session, the observer writes a narrative or running log describing the target student’s behavior. In the last 5 seconds of each 1-minute interval, the observer also records whether the target child is on-task or off-task. On-task versus off-task is determined by the predominant activity sampling method wherein behavior must occur for more than half of the 5-second sampling interval. Hence, the DOF requires that the observer write a narrative and observe on- and off-task behavior simultaneously.

At the end of each 10-minute observation session the observer uses the DOF form to rate the student’s behavior on 97 problem items. Problem items are scored on a 4-point Likert scale: 0 = no occurrence; 1 = slight or ambiguous occurrence; 2 = definite occurrence with mild to moderate intensity and less than 3-minutes duration; and 3 = definite occurrence

with severe intensity or greater than 3-minutes duration. Problem items are short (e.g., “acts too young for age,” “sulks,” “nervous, high strung, or tense”) with 72 items corresponding to items of the Child Behavior Checklist for Ages 6 to 18 (CBCL; Achenbach & Rescorla, 2001) and 83 items corresponding to the Teacher Report Form (TRF; Achenbach & Rescorla, 2001). Factor analyses of data from 212 clinically referred children between 5 and 14 years of age generated six syndrome scales (Withdrawn-Inattentive, Nervous-Obsessive, Depressed, Hyperactive, Attention Demanding, Aggressive; Achenbach & Rescorla, 2001), plus Internalizing and Externalizing scales. The DOF also provides a Total Problem score that is the sum of the 0 to 3 ratings on the 97 items and an on-task score ranging from 1 to 10. In addition, the developers of the DOF recommend observing two comparison children in the same setting (one observed before and one after the target student). The DOF scoring profile provides raw scores for the six syndrome scales, plus *T* scores for Internalizing, Externalizing, and Total Problems. The DOF profile compares scores for the target child (and control children) to a normative sample of 287 children from Nebraska, Oregon, and Vermont.

Psychometric properties. In several studies Pearson correlations were indicative of good interobserver agreement. Averaging across four studies of children in public school classrooms and a residential treatment center (Achenbach & Edelbrock, 1983; McConaughy, Achenbach, & Gent, 1988; McConaughy, Kay, & Fitzgerald, 1998, 1999), mean interobserver agreement was .90 for DOF Total Behavior Problems and .84 for on-task. In an examination of the generalizability of the DOF, Reed and Edelbrock (1983) found that DOF Total Behavior Problems (mean intraclass correlation = .85), but not on-task (mean intraclass correlation = .58), generalized well from one observer to another for individual 10-minute observation sessions. When data from six sessions were combined interclass correlations improved for on-task (mean intraclass correlation = .71), and Total Behavior Problems remained stable (mean intraclass correlation = .86).

The convergent validity of the DOF is supported by significant correlations ($r_s = .37$ to $.51$) between total behavior problem scales of the DOF and TRF (Achenbach & Edelbrock, 1986; Reed & Edelbrock, 1983). The Total Behavior Problems score and the on-task score of the DOF have also been shown to discriminate between boys referred for problem behavior and a sample of typically developing boys matched for age, grade, and race (Reed & Edelbrock, 1983). The treatment sensitivity of the on-task, internalizing, nervous/obsessive, and depressed scales (McConaughy et al., 1999) has been demonstrated in evaluations of school-based programs to prevent emotional disturbance (McConaughy et al., 1998, 1999).

State-Event Classroom Observation System (SECOS)

The SECOS (Saudargas, 1997) was designed to quantify student behavior as part of a comprehensive multimethod assessment and to assess the effectiveness of classroom interventions. It has been used in research studies involving students from first grade through high school. Learning the code typically requires 13 to 15 hours of training (Saudargas, 1997). For the SECOS, momentary time sampling is used to derive an estimate of the amount of time the student engages in the following six "state" behaviors: (a) school work (e.g., a student is solving a math problem in a workbook), (b) out of seat (e.g., student leaves seat without permission), (c) looking around (e.g., student looks out window), (d) social interaction with child (e.g., student talks to neighbor about school work), (e) social interaction with teacher (e.g., teacher is helping student solve a math problem), and (f) other activity (e.g., sharpening pencil). The frequency of five additional "event" behaviors are recorded in 15-second intervals: (a) raise hand (e.g., student raises hand in response to a teacher question), (b) calling out to teacher (e.g., student calls teacher to ask for help), (c) approach child (e.g., student taps neighbor on the shoulder), (d) other child approach (another child taps the target student on the shoulder), (e) teacher approach (e.g., teacher asks student a question). Out of seat appears as both a state

and an event category, which allows for an estimate of both the frequency and the duration of this behavior.

The author of the SECOS recommends observing a classroom peer for comparison purposes. Although no guidelines are offered to direct the collection of such data in the SECOS manual, research studies have collected target and peer data in alternating 20-minute sessions (cf., Slate & Saudargas, 1986a). Also, normative data for the SECOS are available for children in first through fifth grade. The normative sample consisted of 500 children from 10 schools in East Tennessee. Due to a lack of statistically significant differences in scores between boys and girls in the normative sample, these data are grouped together in *T*-score conversion tables by grade.

Psychometric properties. Interobserver agreement using the SECOS appears good. Fellers and Saudargas (1987) reported an average total agreement of $.81$. In another study, Slate and Saudargas (1986a) found total agreement to range from $.75$ to 1.0 for 25% of observations.

Although there do not appear to be any published data supporting the convergent validity of the SECOS, two studies have examined the accuracy of the SECOS. Saudargas and Lentz (1986) found that the association between state and event category scores on the SECOS and real-time recording of the same behaviors on hand-held computers supported the sampling methods employed ($r_s = .67$ to $.92$), as did *t*-tests comparing levels of estimated and real time scores. However, in a later study, Saudargas and Zanolli (1990) found that momentary time sampling in 15-second intervals may not be sensitive to behaviors of short duration (e.g., teacher interactions, verbalizations). If these behaviors are of particular interest, these authors have suggested that shortening intervals (e.g., 5-second) would improve sensitivity, but perhaps at the cost of reliability.

The SECOS significantly discriminated between typically developing children and those with behavior disorders (Slate & Saudargas, 1986a), and learning disabilities (e.g., Fellers & Saudargas, 1987). However, it

should be noted that only observed teacher behaviors and a combination of observed teacher and child behaviors were able to discriminate between boys with learning disabilities and their typically developing same gender peers (Slate & Saudargas, 1986b).

Student Observation System (SOS)

The SOS (Reynolds & Kamphaus, 2004) was designed to assess a broad array of both adaptive and maladaptive classroom behaviors, and is a component of the Behavior Assessment System for Children-2nd Edition (BASC-2; Reynolds & Kamphaus, 2004). It has been suggested that training for the SOS can be accomplished in a 30-minute workshop or by simply reading the manual, but no criterion for training has been reported (Lett & Kamphaus, 1997). The length of observation sessions is typically 30 minutes, but the authors recommend observing the target child across 3 or 4 days in different classrooms to enhance the reliability of measurement.

Using the SOS, the observer takes notes concerning child and teacher behaviors for 27 seconds of each 30-second interval. In the last 3 seconds of each interval, the observer uses a 3-second momentary time sampling procedure to record adaptive and/or maladaptive behaviors exhibited by the target child. The adaptive behaviors are coded using the following four categories: (a) response to teacher/lesson (e.g., answers teacher's question appropriately), (b) peer interaction (e.g., participates appropriately in small group discussion), (c) work on school subjects (e.g., completing a math worksheet alone), and (d) transition movement (e.g., walking to blackboard when asked to do so). The following nine categories are grouped together as maladaptive behaviors: (a) inappropriate movement (e.g., walking around classroom when inappropriate), (b) inattention (e.g., doodling on book), (c) inappropriate vocalization (e.g., teases another student), (d) somatization (e.g., complains about a headache), (e) repetitive motor movements (e.g., plays with hair), (f) aggression (e.g., intentionally breaks a neighbor's pencil), (g) self-injurious behavior (e.g., pulls own hair), (h) inappropriate sexual behavior (e.g., strokes

self), and (i) bowel/bladder problems (e.g., wets pants). At the end of the 30-minute observation session, the observer then reviews notes and rates the student's behavior on 65 behavior items on a 3-point Likert scale (NO = *never observed*, SO = *sometimes observed*, FO = *frequently observed*). Items are grouped according to the aforementioned adaptive and problem behavior categories. For the problem behavior items, there is a column to indicate whether the behavior was disruptive to the class.

Psychometric properties. Unfortunately, little published data are available concerning the psychometric properties of the SOS. The manual for the BASC-2 reports no data concerning technical adequacy of the SOS. In one study, the SOS was evaluated with regard to its ability to discriminate a group of 37 children with ADHD from a group of 18 typically developing children (Lett & Kamphaus, 1997). In this study interobserver agreement was examined in a subsample of participants ($n = 44$) with coefficients reported to be in the .80s. However, the range of interobserver agreement coefficients and the method employed to evaluate interobserver agreement was not reported, making interpretation of these data difficult. Nevertheless, scores on the category of inappropriate movement and the problem behavior composite (of which the inappropriate movement category is a contributor) from the momentary time sampling portion of the SOS significantly discriminated children with ADHD from typically developing children.

Discussion

The purpose of this article was to critically evaluate seven observation systems designed to assess a student's classroom behavior. Table 3 provides a summary of the strengths, limitations, and recommended uses for each of the codes included in this review.

Recommendations for selection of observation codes. Three of the codes, the AET-SSBD, DOF, and SOS, were developed in conjunction with other measures (the SSBD, ASEBA, and BASC-2, respectively) and are

closely aligned with the constructs assessed by these measures. The remaining four codes were developed to assess key behavioral domains, although some focus exclusively on problem behaviors (i.e., ADHD-SOC, COC), whereas others (i.e., BOSS, SECOS) focus on positive behaviors (e.g., academic engagement) as well as problem behaviors. As such, the primary target behavior(s) of interest will be one of the initial factors guiding the selection of a potential observation code from among those included in this review.

Beyond the consideration of target behaviors, reliability and validity evidence must be weighed when deciding which observation code to use. With the exception of the SOS, all of the reviewed codes have minimally sufficient reliability evidence. With regard to validity, all of the codes have at least some evidence to suggest that scores differentiate between students with classroom behavior difficulties and students without such difficulties. Only three of the codes (AET-SSBD, ADHD-SOC, and DOF), however, have published evidence of convergent validity. Similarly, only four of the codes (ADHD-SOC, COC, DOF, and BOSS) have published evidence to support their use for monitoring change in classroom behavior in response to intervention.

Given the strengths and limitations of the available data, six of the codes have sufficient evidence *to be used as part of a multimethod assessment*. Based on the available data, the ADHD-SOC and DOF appear to have the most support for use in the multimethod assessment of externalizing problems, and the DOF is the only code appropriate for assessing internalizing problems. The COC shows promise in the assessment of classroom behaviors associated with ADHD, whereas the SECOS, BOSS, and AET-SSBD demonstrate promise in the assessment of positive behaviors in the classroom setting. The extremely limited published evidence available for the SOS precludes its use at the current time.

Recommendations for observation best practices. In the beginning of this article we listed the threats to the validity of observational assessment identified by Merrell (1999), including the use of poorly defined behavior

categories and inappropriate code selection. By presenting information concerning the seven codes reviewed here we hope to enhance the validity of observations by facilitating the appropriate selection of well-validated coding schemes for particular assessment tasks. There are other strategies that observers can use to maximize the validity of observational assessments. First, it is incumbent on the observer to ensure that they are adequately trained on a given code and that the consistency of their observations does not decline over time. Training requirements (summarized in Table 1) should be taken into consideration when selecting any given code. One way to ensure the adequacy of training is to utilize a precoded videotape to determine if a minimum degree of accuracy has been achieved. Unfortunately, such tapes are available only for the AET-SSBD. Alternatively, observers can check their agreement with a second observer (see Hintze, 2005, for methods to calculate interobserver reliability). In addition to ascertaining whether observers have been trained to criterion initially, it is also necessary periodically to check reliability to curb observer drift.

Second, several observations are necessary to achieve a reliable estimate of a student behavior (see Hintze, 2005). Although this has been discussed in terms of traditional measurement theory (e.g., the measurement of a trait), it would seem equally valid for behavioral approaches to assessment wherein one is more interested in assessing differences in a given behavior across conditions. As such, it is recommended that if one wishes to make comparisons of student behavior across settings, multiple observations should be performed within each setting. Third, the normative data that are currently available (e.g., AET-SSBD, SECOS) appear inadequate due to either sampling techniques, the age of the data, or both. Further, given the variability in ecology across educational settings (e.g., task demands, classroom rules, classroom management skills, quality of instruction) standardized norms seem ill suited for observational assessment. Hence, it is recommended that local normative data be collected for frequently used codes and that

for each assessment, data be collected on one or more peers under the same conditions as the target child.

Two final considerations for ensuring observation best practices are reactivity and observer bias. Reactivity refers to a target child altering behaviors as a result of being observed, resulting in inaccurate estimates of actual target behaviors. One strategy for minimizing reactivity is to conduct multiple observations to increase the child's familiarity/comfort with the observer in the classroom. Observer bias also affects observation accuracy and refers to the tendency of an observer to consistently view (and record) observed behaviors in a particular way (e.g., negative, positive). Adequate training and periodic reliability checks described previously are perhaps the most effective way to minimize the likelihood of observer bias.

Finally, no assessment should rely on a single measurement method, particularly when reliability and validity evidence is limited. Assessments are enhanced when multiple methods are employed (Cone, 1978) to assess behavior across multiple dimensions (Achenbach, 1993). Hence, observations, like any other assessment methodology, should only be used as part of a broader assessment battery irrespective of the assessment domain.

Future Research Directions

There are multiple critical directions for future research to ensure identification of appropriate (and inappropriate) uses of the standardized behavior observation codes reviewed herein. In light of the limited number of studies evaluating convergent validity, each of the codes would benefit from additional studies addressing this type of evidence. Second, evidence for treatment sensitivity is nonexistent for some codes and minimally sufficient for others. Studies examining treatment sensitivity are essential if these systems are to be used to evaluate intervention effectiveness.

One final line of validity evidence not currently addressed in studies of these codes is the representativeness of observed behavior based on a small number of observations. Despite the common professional belief that re-

sults of observation are the "gold standard" in the assessment of behavior, studies (e.g., Doll & Elliott, 1994; Hintze & Matthews, 2004) have raised important questions regarding the validity of a small number of observations to measure classroom behavior. Given that most practitioners rarely have time to engage in a large number of observations for an individual student, determining the validity of a single observation (or small number of observations) with each of these codes is essential for justification of their use in professional practice for screening and diagnosis.

Conclusions

The direct assessment of student behavior has been a critical component of comprehensive evaluations of student behavior in classroom settings. The seven observation codes reviewed in this article have been developed to provide practitioners with a standardized framework for measuring classroom behavior. With the exception of one code, all have published interobserver agreement evidence to support their use with school-age populations. Most also have some evidence of predictive validity and treatment sensitivity, though much of this evidence is limited to single studies with samples that are small to moderate in size. Even less evidence is available related to convergent validity. As a result of these limitations in existing evidence, much research is necessary to ensure that these codes are used for appropriate assessment purposes, behaviors, and target students. Until the completion of such studies, practitioners are encouraged to select measures cautiously and use multiple methods in screening, diagnosis, and evaluation of treatment effectiveness.

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